Perpendicular bisectors

1

a $y=-2x+2$

b $x+2y+7=0$

c $x+4y-29=0$

d $4x-3y+11=0$

2

a

i $(3,-1)$

ii $m=-1$

iii $y=x-4$

b

i $(-1,-2)$

ii $m=-\frac{2}{3}$

iii $3x-2y-1=0$

c

i $(0.5,0.5)$

ii $m=\frac{3}{7}$

iii $7x+3y-5=0$

d

i $(-1,1)$

ii $m=\frac{4}{7}$

iii $7x+4y+3=0$

3

a $x+7y-9=0$

b $x+4y-17=0$

c $x-5y+60=0$

d $5x+6y-79=0$

e $3x+7y-14=0$

f $3x+y-2=0$

g $5x-2y-21=0$

h $x+2y-100=0$

4 $5x-2y+9=0$

Applications

5

a $y=x$

b Yes, both points lie on the perpendicular bisector.

6

a

i $2x + 3y - 39 = 0$

ii $3x - 5y + 2 = 0$

iii $5x - 2y - 37 = 0$

b $(19.9, 6.37)$